

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: Rescue Rats

Every Line/Maze team has to submit a bill of materials for their

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

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#	Component	Part name	Autor	Source	Quantity	Unit cost
1	Microcontroller	Raspberry pi 5	Raspberry Pi Foundati	Rpishop	1	184,11 €
2	Microcontroller	ESP32-S3 Microcontroller, 2.4GHz Wi-	Espressif Systems	Elecom.sk	1	12,42 €
3	cameras	pi cam 3 wide	Raspberry Pi Foundati	Rpishop	2	40,05 €
4	LED	LED RGB 5 mm		Techfun.sk	1	0,05 €
5	Connecting parts	screws, cables		Local store	1	25,00 €
6	Battery	KAVAN LI-PO 2700MAH/14,8V	Kavan	Kavan	2	39,30 €
7	Motor driver	MDD10A	Cytron Technologies	neven.sk	2	27,90 €
8	RGB sensor	TCS34725 RGB Sensor	Adafruit	Techfun.sk	1	10,15 €
9	Microcontroller Acc	ESP32-S3-DevKitC-1 Extended version b	Generic	Techfun.sk	1	3,31 €
10	Sensor	VL53L1X Laser Ranging Sensor Module	STMicroelectronics	Techfun.sk	8	10,90 €
11	Sensor Cover	VL53L1X Laser Ranging Sensor Lens Co	Generic	Techfun.sk	8	0,50 €
12	Wheels	Mecanum Wheels 60mm	Generic	AliExpress	4	2,32 €
13	Gyroskop	BNO055 sensor modul MCU 9DOF	Adafruit	Techfun.sk	1	19,95 €
14	Motor	JGY-370 DC Motors	Generic	AliExpress	4	6,70 €
15	Step/down converter	Step-down BUCK Converter xy 3606	Generic	Elecom.sk	1	1,30 €
16	Servo	Servomotor SG90 - 360°	TowerPro	drotik elektro	1	2,54 €
17	Switch	Switch ON/OFF E-TEN 1021	Generic	Techfun.sk	1	3,15 €
18	Robot shell	Custom made	Karolína, Martin	3D printer	1	30€
19						
20						
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23						

Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Euro EUR	633,76 €	\$733,44

Unit cost (US	Total Cost	Total Cost
\$213,07	184,11 €	\$213,07
\$14,37	12,42 €	\$14,37
\$46,35	80,10 €	\$92,70
\$0,06	0,05 €	\$0,06
\$28,93	25,00 €	\$28,93
\$45,48	78,60 €	\$90,96
\$32,29	55,80 €	\$64,58
\$11,75	10,15 €	\$11,75
\$3,83	3,31 €	\$3,83
\$12,61	87,20 €	\$100,92
\$0,58	4,00 €	\$4,63
\$2,68	9,28 €	\$10,74
\$23,09	19,95 €	\$23,09
\$7,75	26,80 €	\$31,02
\$1,50	1,30 €	\$1,50
\$2,94	2,54 €	\$2,94
\$3,65	3,15 €	\$3,65
\$34,72	30,00 €	\$34,72
	- €	\$0,00
	- €	\$0,00
	- €	\$0,00
	- €	\$0,00
	- €	\$0,00

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#	Name	Software's Tool/Library	Description	Source
1	Open CV	Library	Image processing library	OpenCV
2	IDE Arduino	Software's Tool	Software	Software Arduino
3	Visual Studio Code	Software's Tool	Software	Visual Studio Code
4	Git	Software's Tool	Version control for managing functions	Git
5	Fusion 360	Software's Tool	3D CAD shell design	Fusion Autodesk
6	VL53L1X	Library	For controlling the VL53L1X distance sensor	https://github.com/pololu/vl53l1x-arduino
7	Adobe Premier Pro 2026	Software's Tool	For video editing	https://www.adobe.com/products/premiere
8	Numpy	Library	Array and math library	NumPy
9	Matplotlib	Library	Python graphics generation library	Matplotlib — Visualización con Python
10	Picamera2	Library	Raspberry Pi camera integration	https://picamera2.com/
11	PySerial	Library	Cross-platform Python library for serial ports	https://www.pyserial.com/
12	BNO055	Library	For controlling Adafruit BNO055 Absolute Orientation Sensor	https://learn.adafruit.com/adafruit-
13	TCS34725	Library	Arduino library for TCS34725 RGB Color Sensor	https://www.arduino-libraries.
14	ESP32Servo	Library	This library provides advanced PWM control for ESP32 Servo	https://docs.arduino.
15	KiCad	Software's Tool	Open source PCB design / electronics CAD	https://www.kicad.org/
16	Bambu Studio	Software's Tool	For printing 3D models	https://bambulab.com/en/download/studio
17				
18				

Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
EUR	0,00 €	\$0,00

Autor	Cost (Local	Cost (USA
Opencv.org	FREE	FREE
Arduino.cc	FREE	FREE
Microsoft	FREE	FREE
Software Freedom	FREE	FREE
Autodesk	FREE/EDUCATION	FREE/EDUCATION
Pololu	FREE	FREE
Adobe	FREE/EDUCATION	FREE/EDUCATION
Numpy developers	FREE	FREE
Matplot lib dev team	FREE	FREE
Picamera2 dev	FREE	FREE
PySerial dev team	FREE	FREE
Adafruit	FREE	FREE
<u>Adafruit</u>	FREE	FREE
Kevin Harrington	FREE	FREE
KiCad dev team	FREE	FREE
Bambu Lab	FREE	FREE